INDIVIDUAL PROJECT

JESSICA AGNESIA TATAUNG

Retail Business Performance Analysis

Python | Power BI

ABOUT ME

Pharmacy graduate with a strong analytical and research background, currently transitioning into Data Analytics through intensive training and hands-on projects. Proficient in Python, SQL, Tableau, Power BI, and Excel, with a passion for transforming data into actionable business insights and supporting data-driven decision-making.

🎓 B.Sc. in Pharmacy, ITERA (GPA 3.48/4.00)
📊 Data Science Bootcamp Graduate
🏅 Google Data Analytics Professional Certificate
💻 Python | SQL | Tableau | Power BI | Excel

Retail Business Performance Analysis

Identifying Profitability Drivers & Strategic Business Recommendations

Global Superstore

51,290 Transactions

2011 - 2014

Python • Pandas • Matplotlib • Seaborn • Power BI

2 Business Context

Retail companies often achieve strong sales growth, but profitability can vary significantly across products, markets, and pricing strategies. Understanding the key drivers of profitability is essential for making better pricing and product decisions.

Business Question

How can the company improve profitability while sustaining long term sales growth?

Business Objectives

Evaluate Business Performance

Analyze sales and profit trends from 2011-2014

Identify Profitability Drivers

Evaluate products, markets, and customers contributing to business performance

Investigate Root Cause

Identify factors associated with declining profitability

Provide Business Recomedations

Deliver actionable recommendations supported by data

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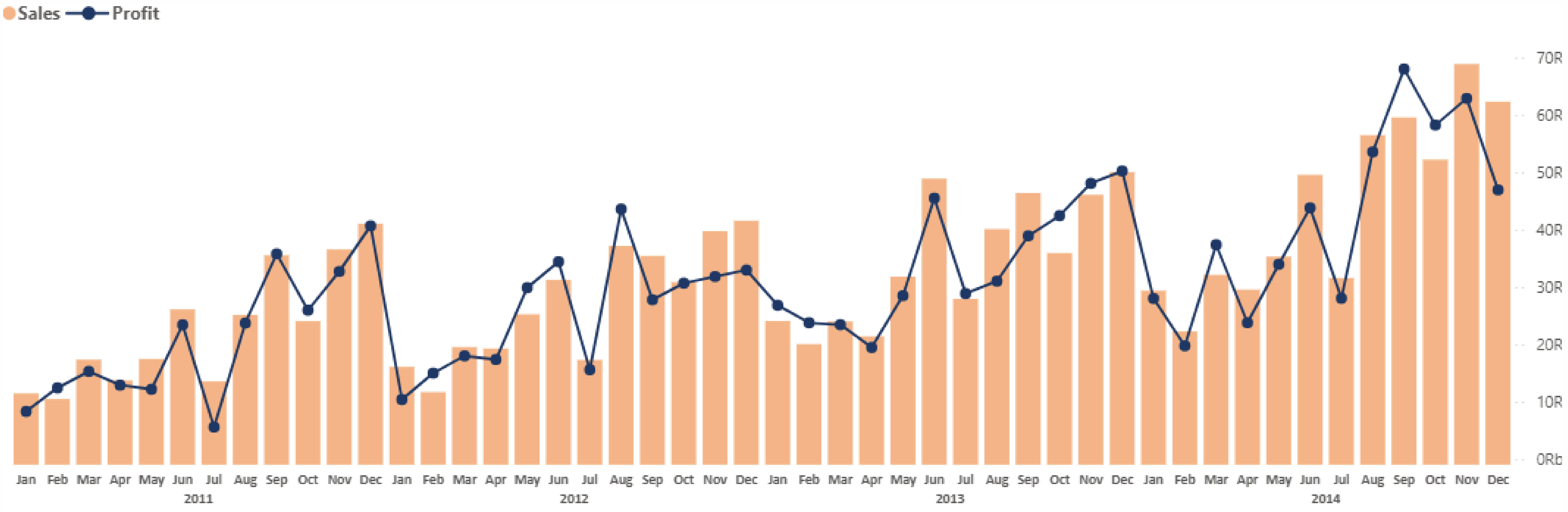
Project Workflow

Business Understanding → Data Cleaning → Business Performance Analysis → Product Performance Analysis

Root Cause Analysis → Market Performance Analysis → Customer Performance Analysis → Business Recommendation

Sales & Profit Metrics (Jan 2011 - Dec 2014)

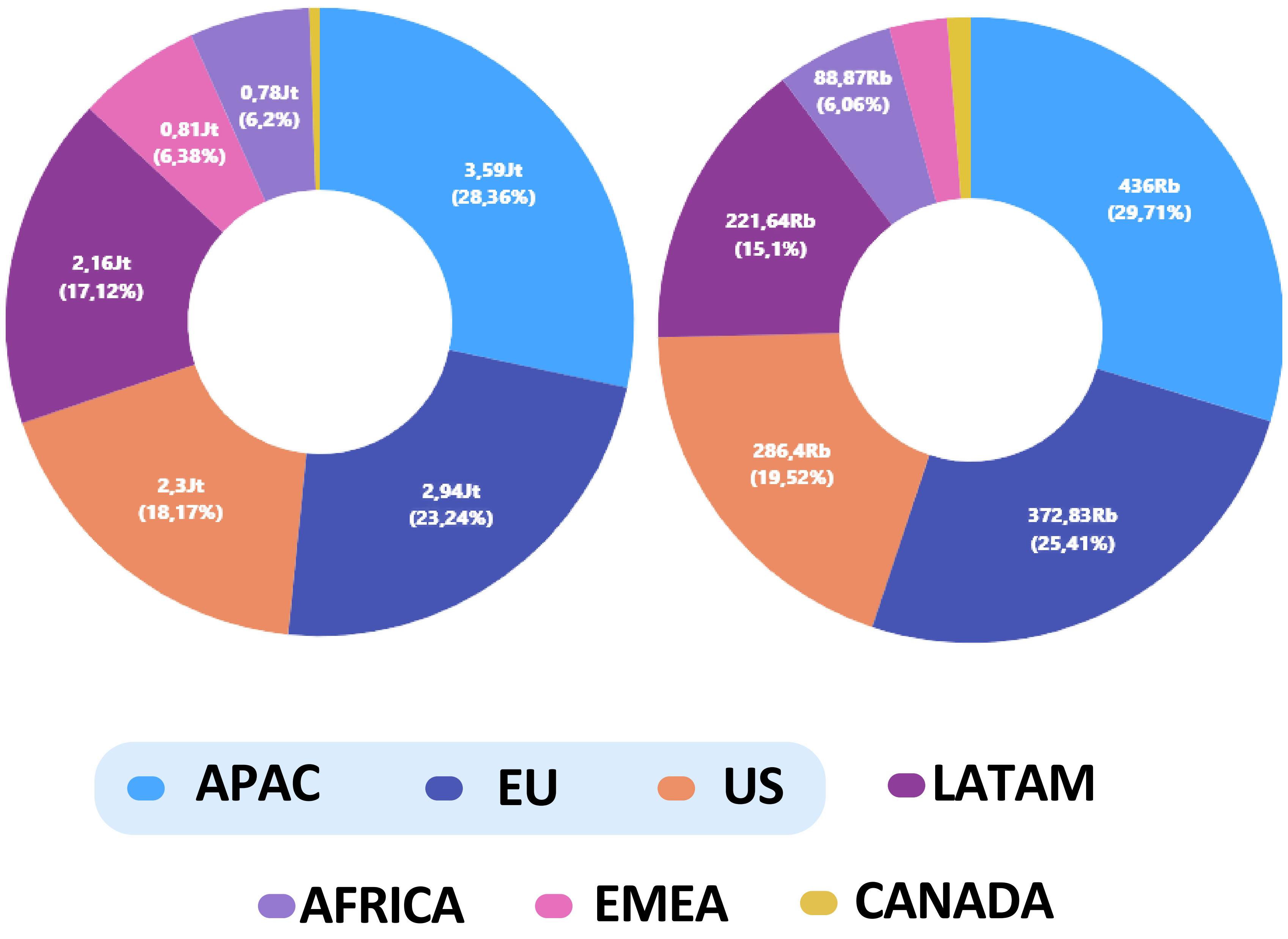


Sales exhibited similar month-to-month patterns across 2011–2014, with the sharpest decline consistently occurring in July, followed by a broad recovery during the second half of the year despite a recurring pullback in October.

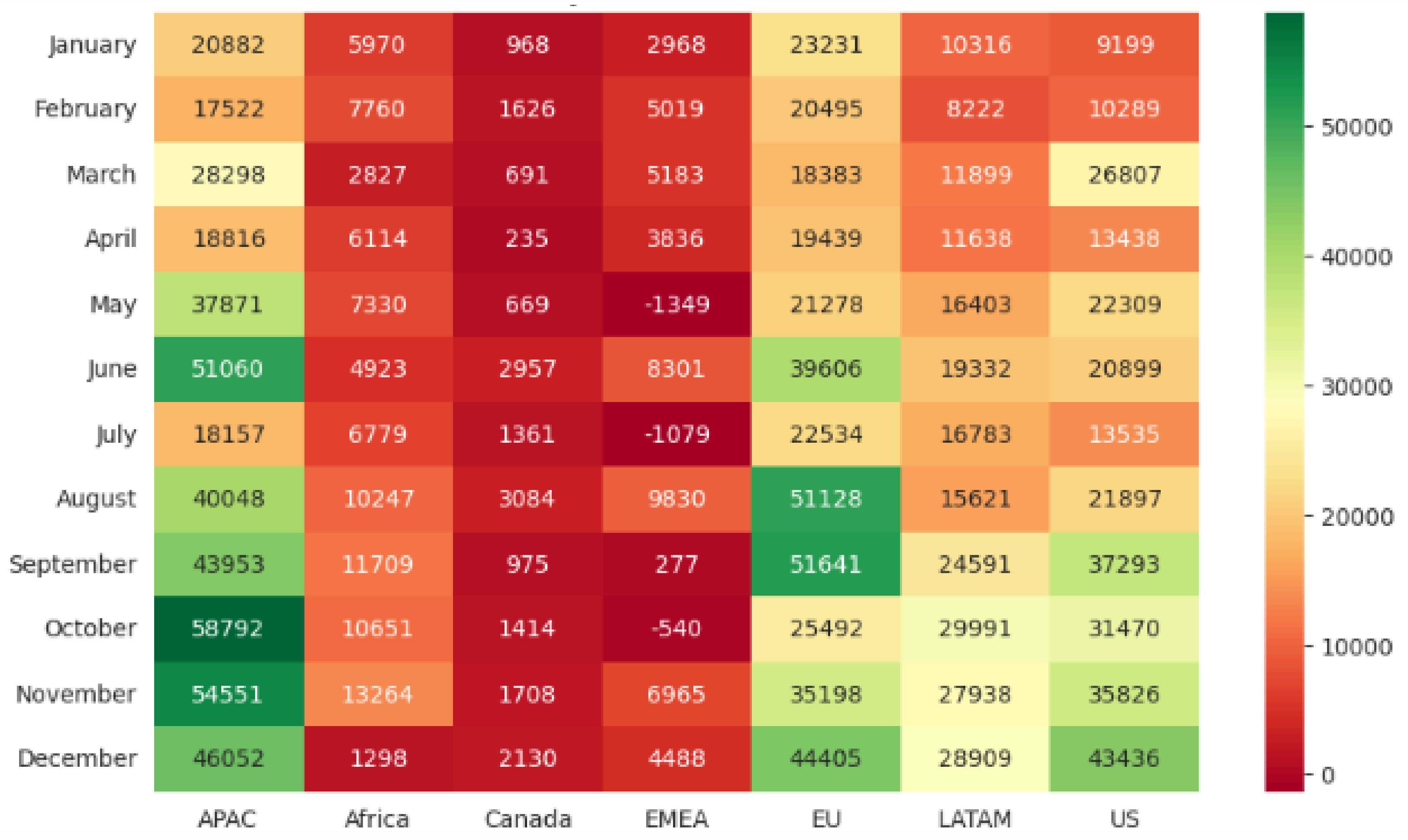
APAC Was the Primary Driver of the July Sales and Profit Decline

APAC Generated the Highest Sales and Profit

Sales Contribution **Profit Contribution**



Profit by Month Vs Market



Year-over-year changes in profitability were primarily driven by the company's top three markets: APAC, EU, and the US.

Why Did APAC Performance Decline in July?

Two main factors drive the profit decline in July: lower demand and weaker profitability

LOWER DEMAND (ORDER VOLUME)

Sharp decrease in order volume across key profit generating sub categories

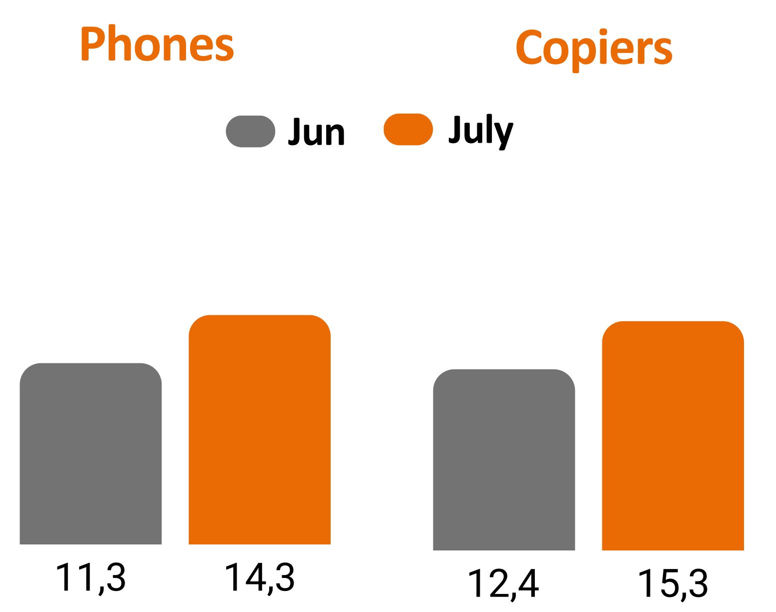
Order Volume Change (Jun - Jul)

Phones	Copiers	Bookcases	Chairs
↓ 44.3%	↓ 52.9%	↓ 42.2%	↓ 61.3%

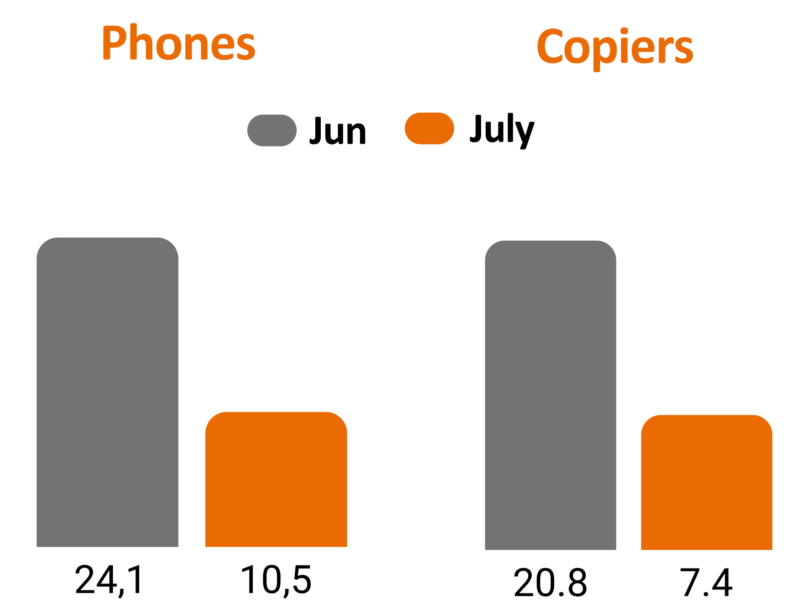
LOWER PROFITABILITY (MARGIN PRESSURE)

Higher discount levels led to lower profit margins, especially in Phones and Copiers

Average Discount (Jun - July)

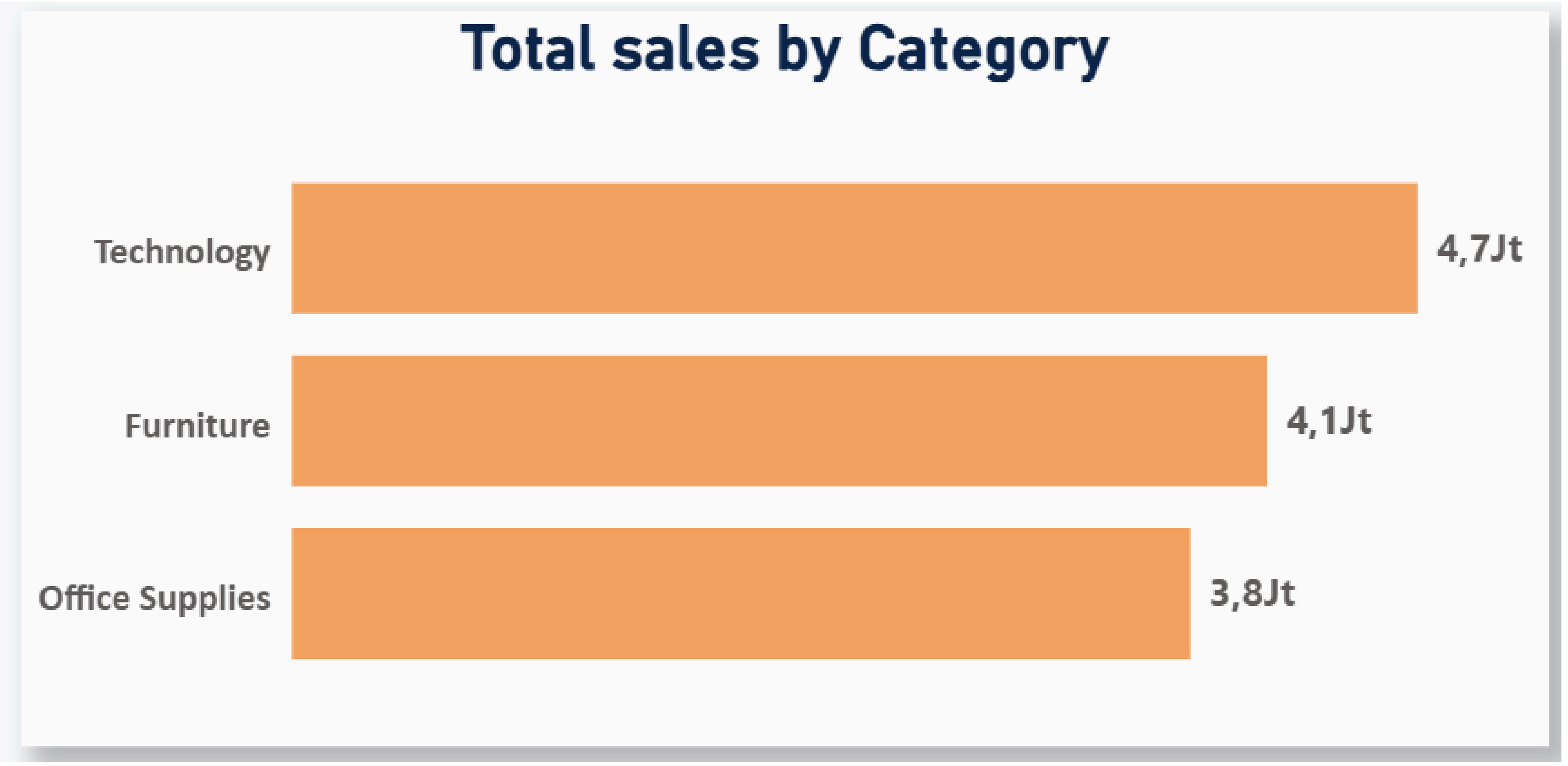


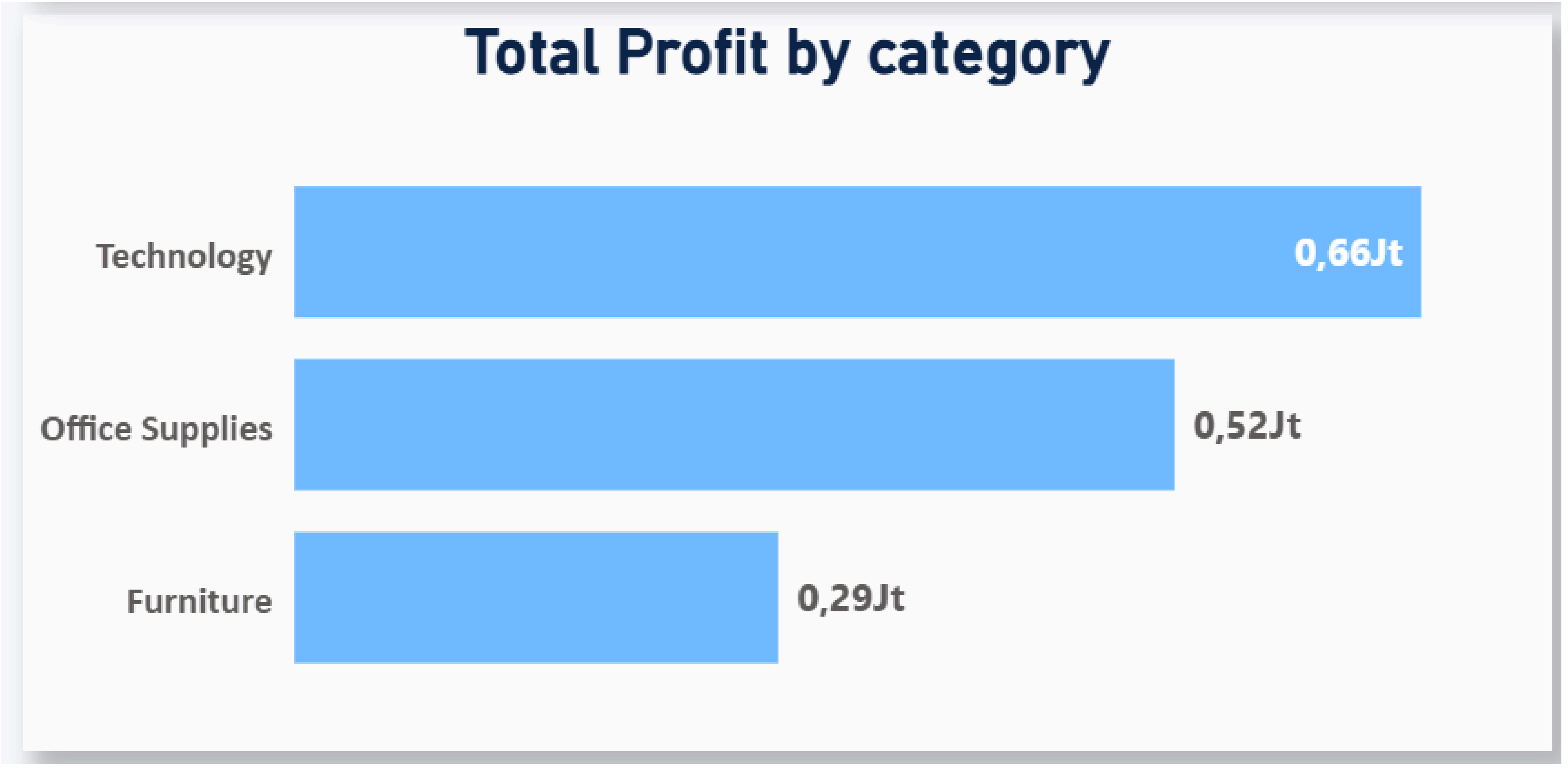
Profit Margin (Jun - July)



Lower demand reduced sales, while higher discounting compressed margins, leading to a significant profit decline in July

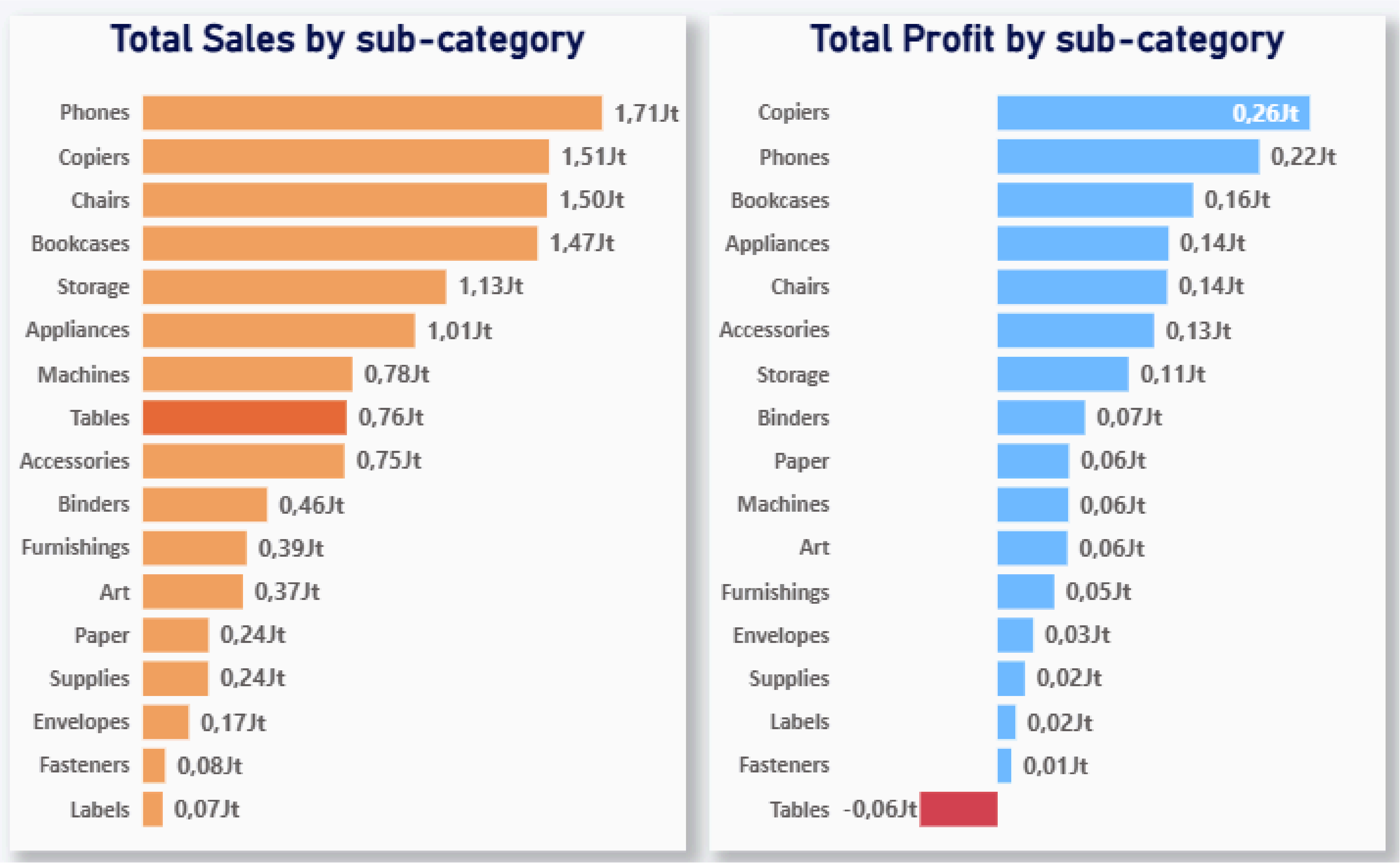
Technology Led Profit Growth,
While Furniture Weighed Down Overall Profitability





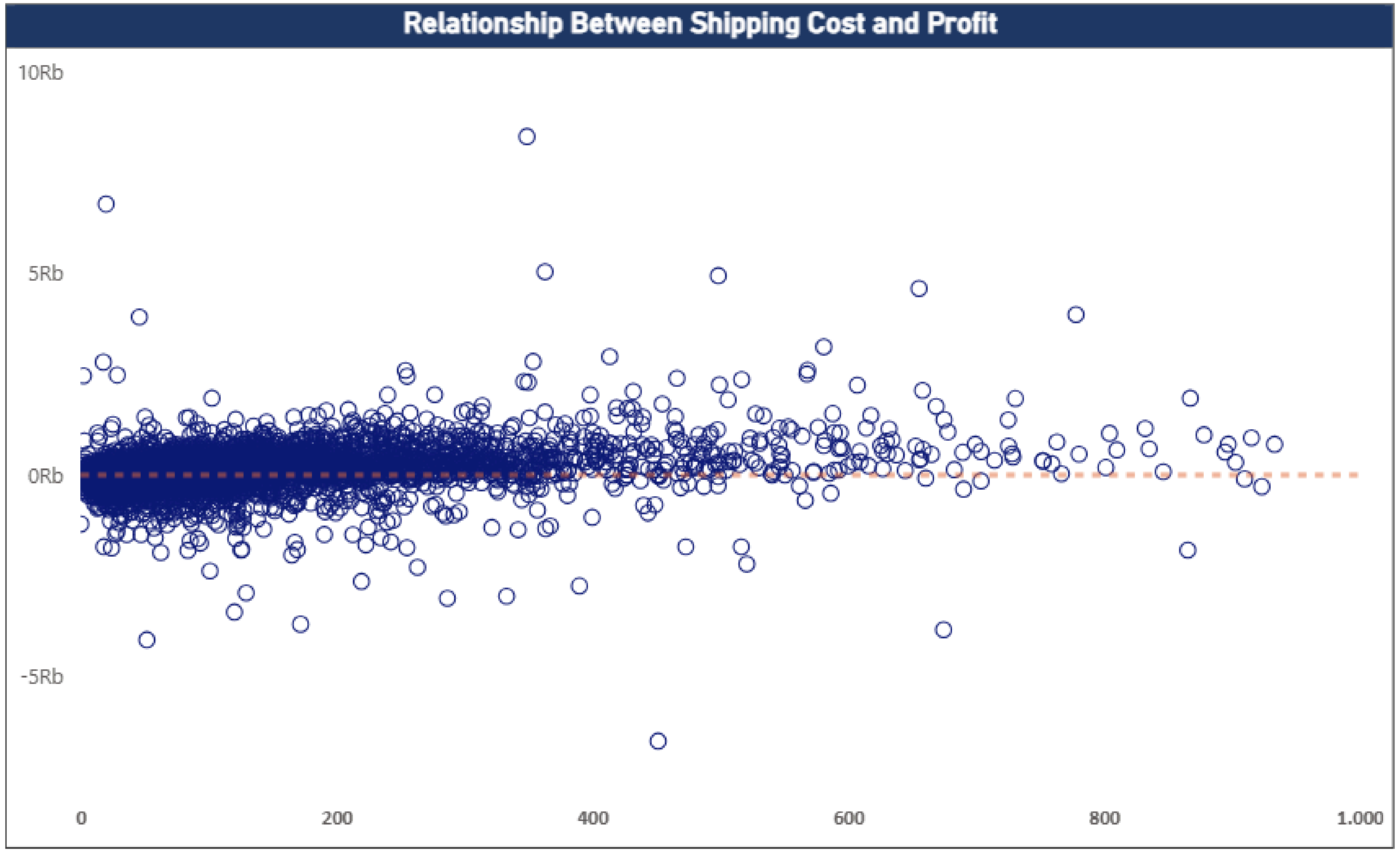
Technology generated the highest sales and profit, while Furniture produced comparable sales but significantly lower profitability, indicating margin inefficiencies within the category.

Strong Sales Did Not Always Translate into Strong Profitability

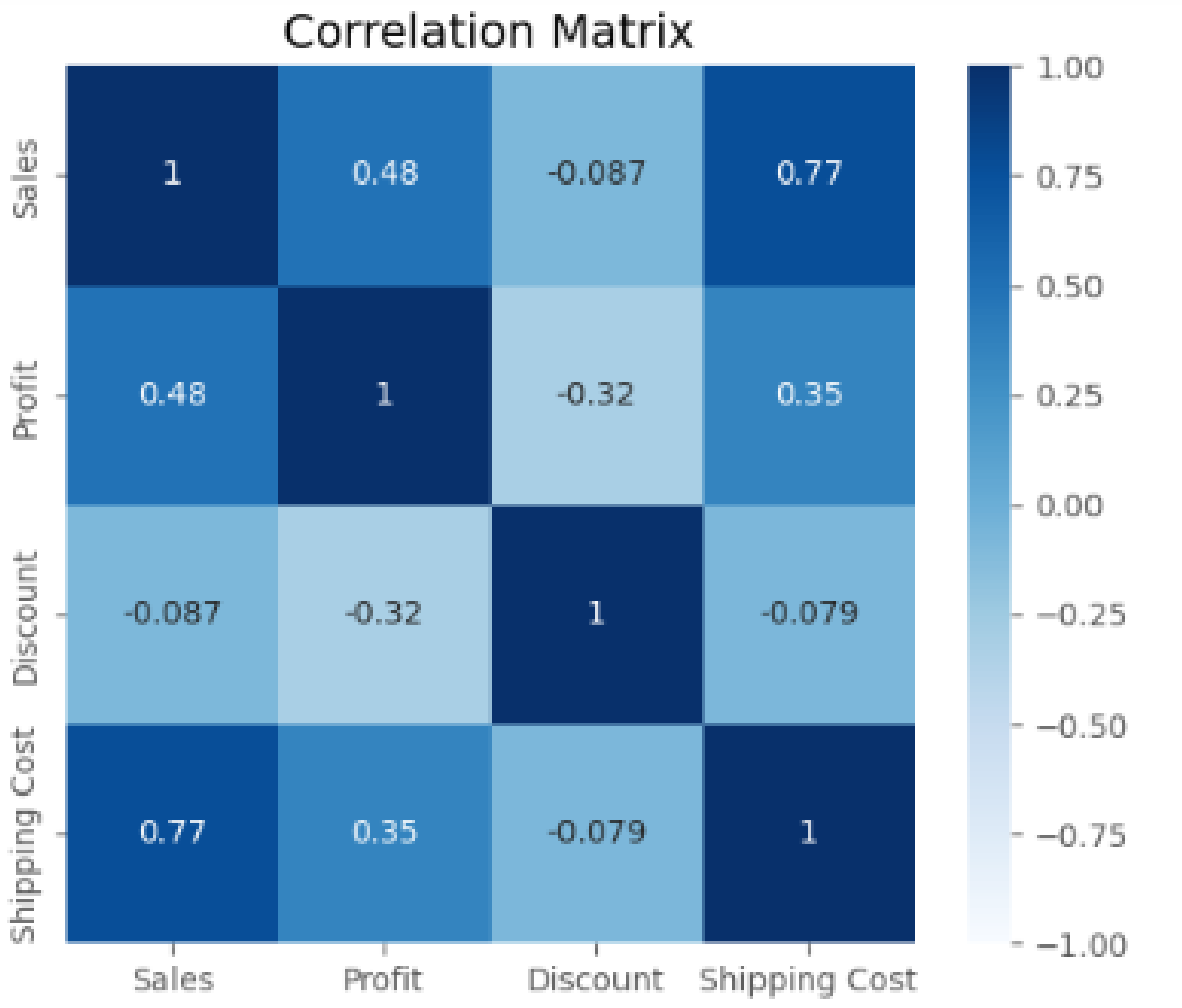


Despite generating approximately $0.8M in sales, Tables was the only sub-category that recorded a negative profit, warranting further investigation into its profitability drivers.

Identifying the Key Driver of Profitability

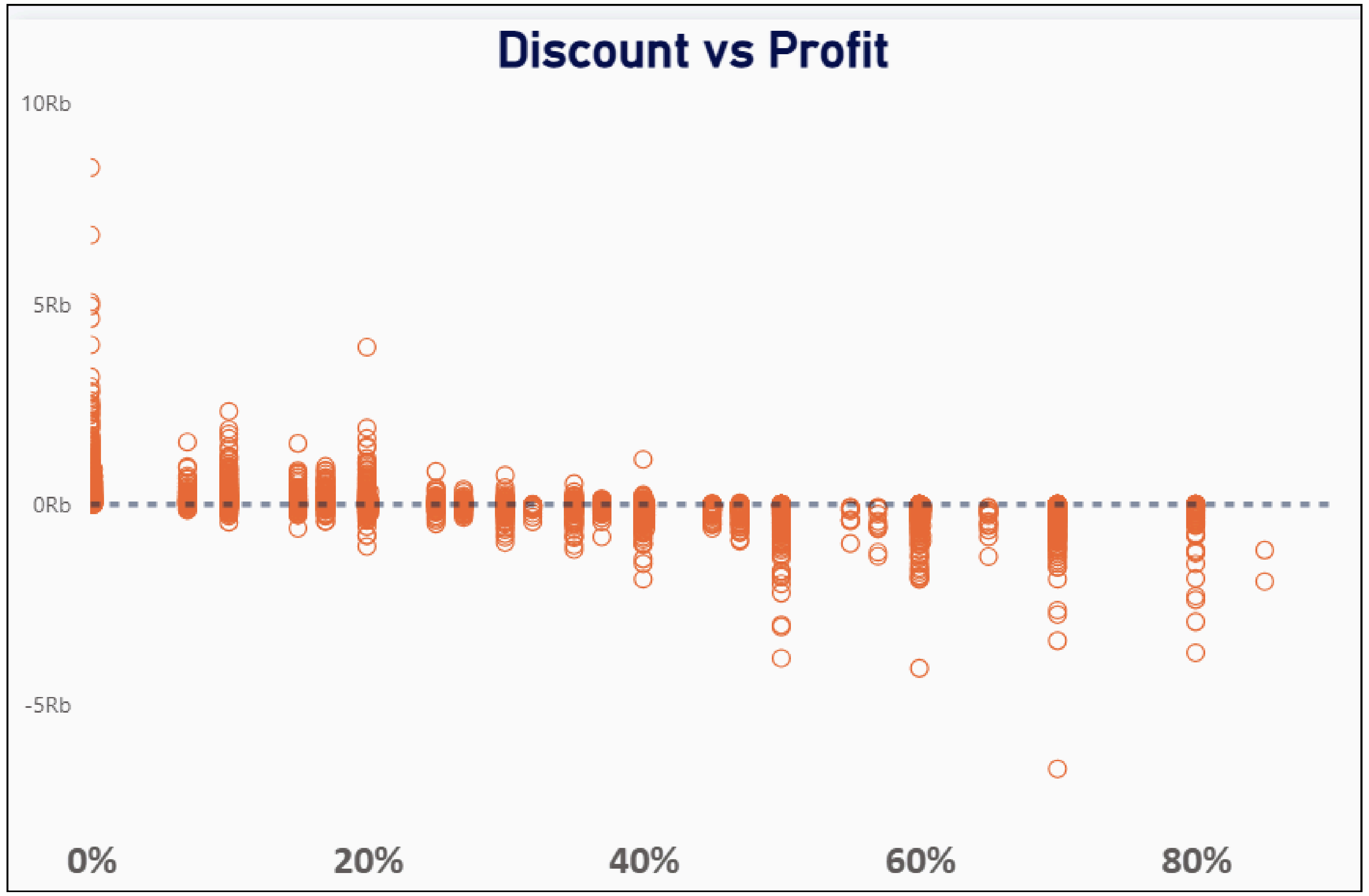


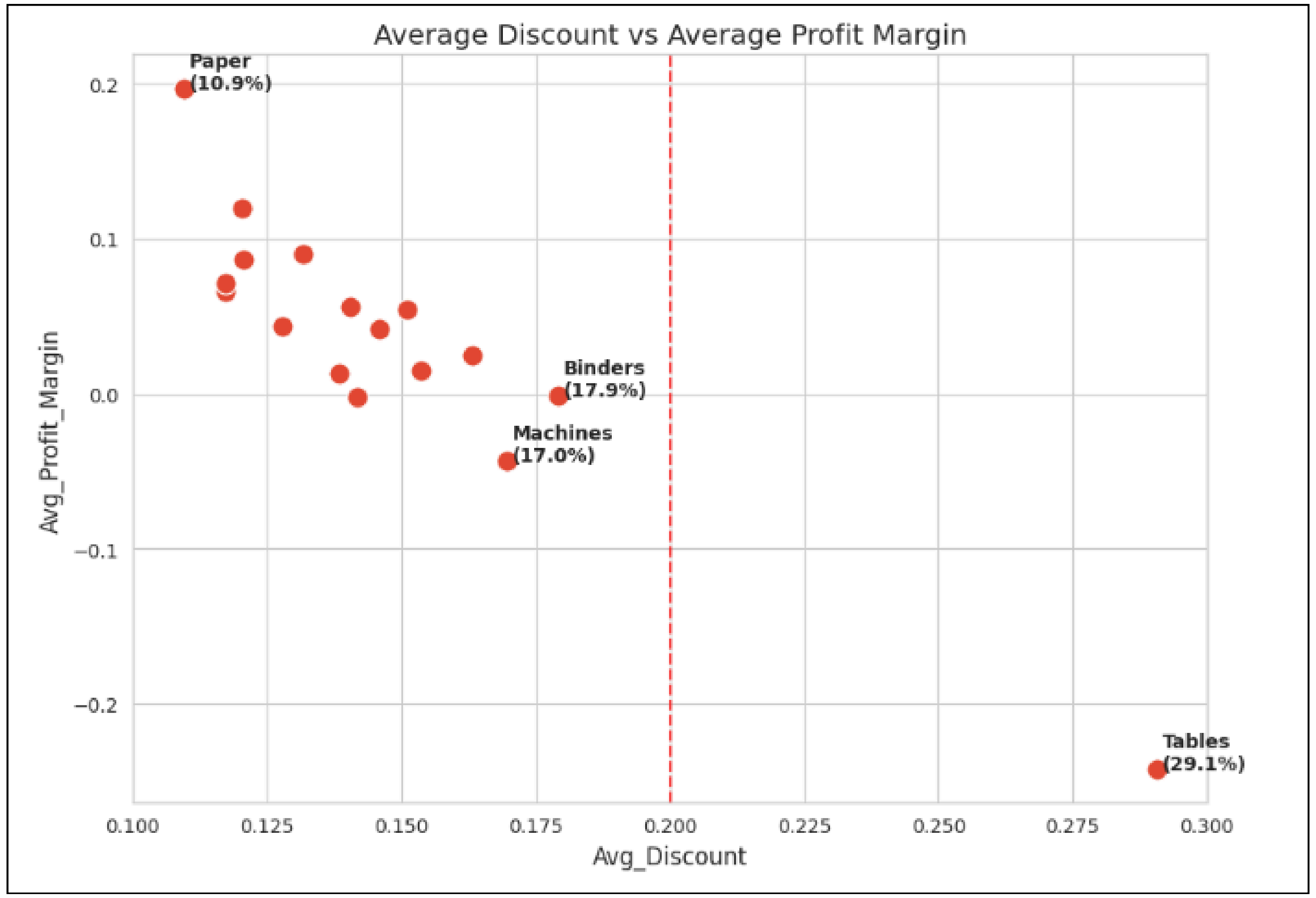
Shipping cost showed a positive correlation with profit (r = 0,35). Shipping cost was not the primary driver behind low profitability



Higher shipping costs were associated with higher profits, indicating shipping cost was not the primary driver of low profitability

Higher Discounts Erode Profitability, with Tables Experiencing the Largest Losses

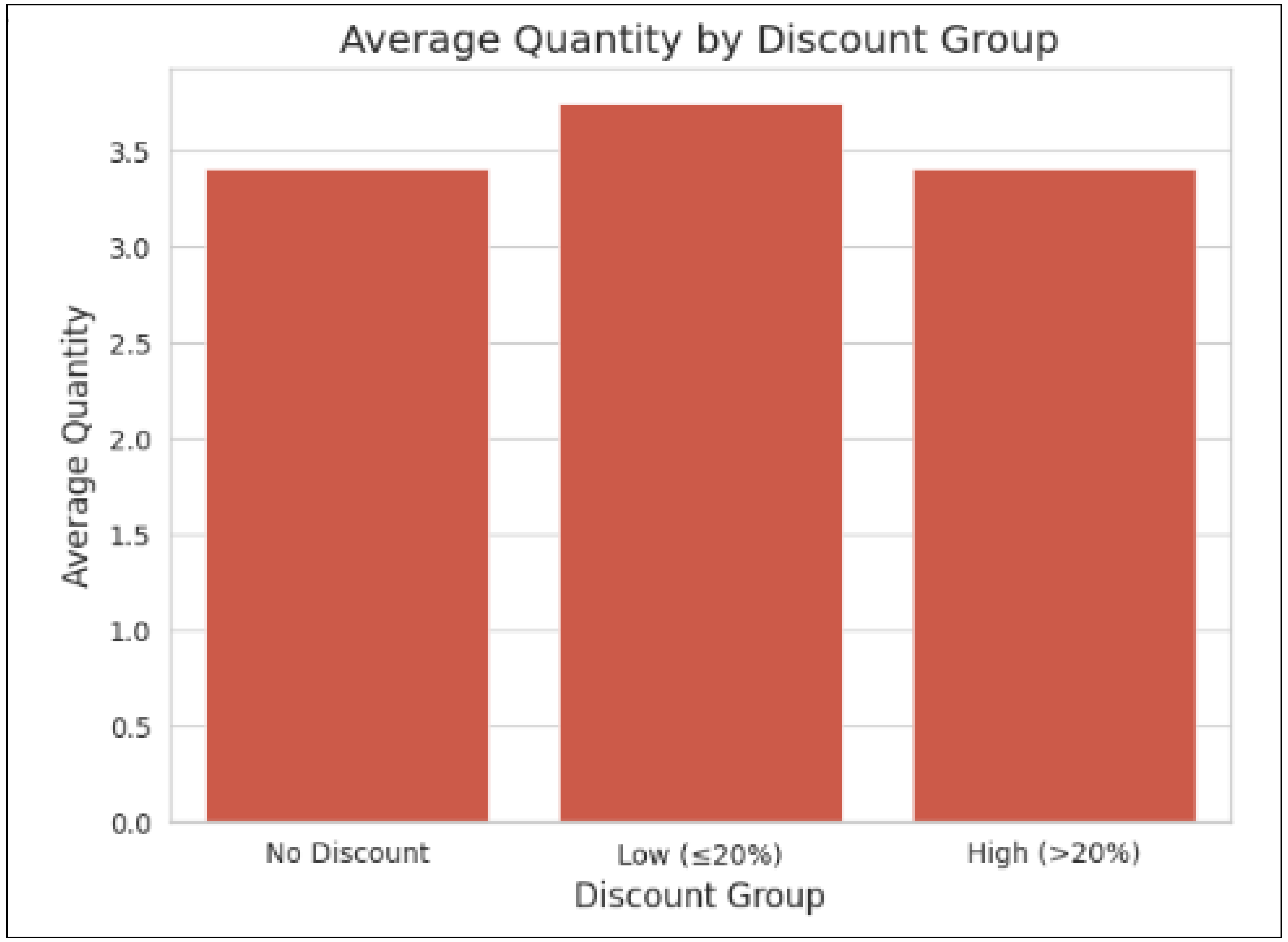




Transactions with discounts above 20% are consistently associated with lower profit margins

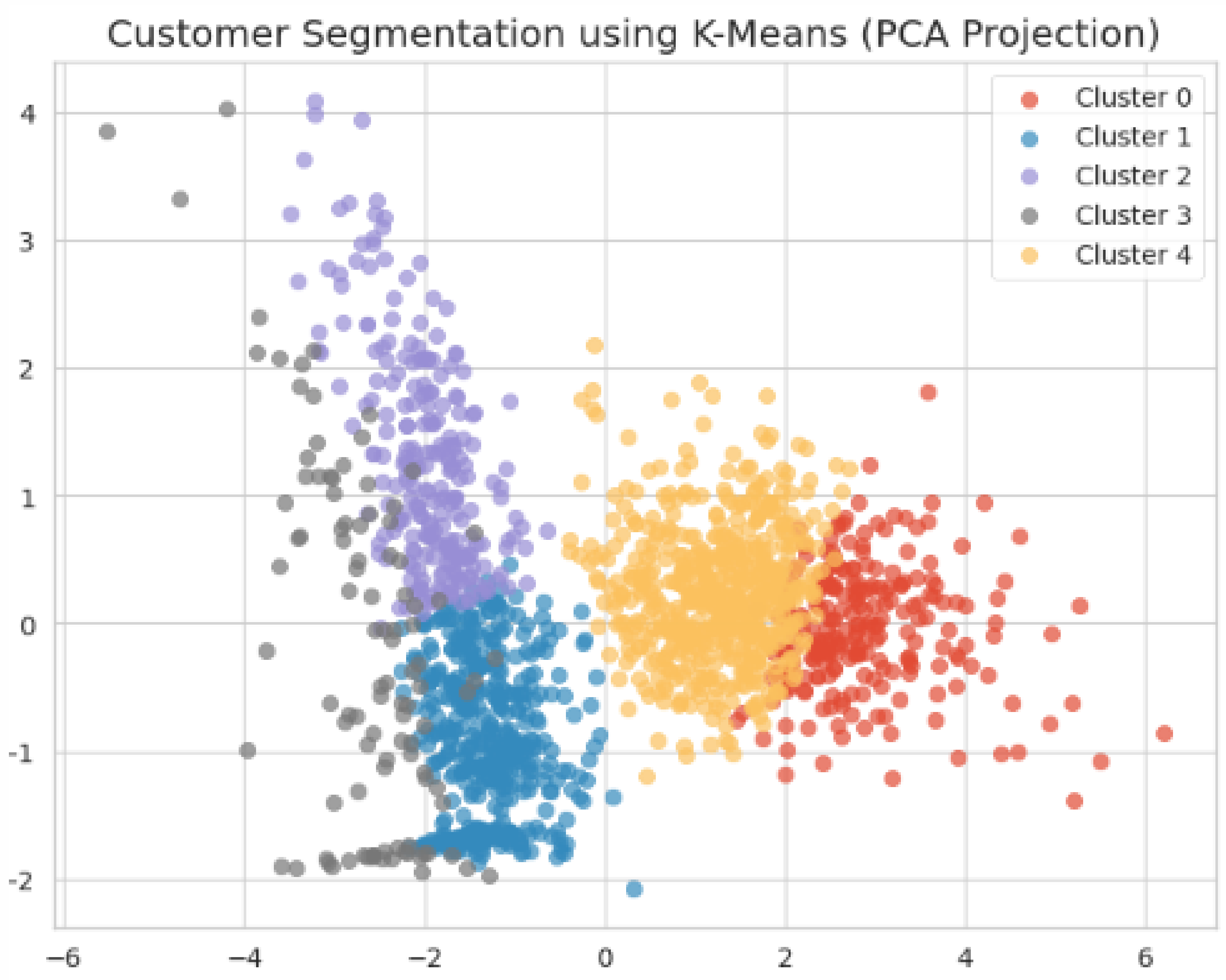
Tables receive the highest average discount (29.1%) and generate the lowest average profit margin (-24%)

Discounts Above 20% Do Not Deliver Additional Sales Volume



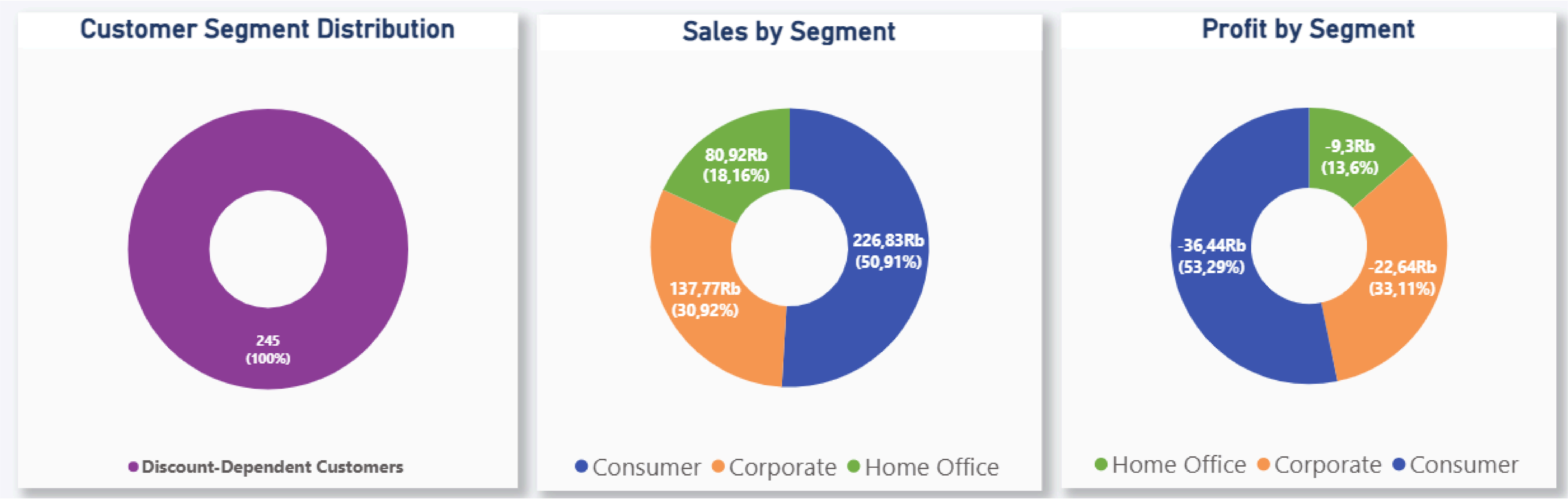
- Discounts up to 20% significantly increase purchase quantity.
- Kruskal–Wallis ($p < 0.05$) and Dunn's post-hoc test confirm that discounts above 20% do not generate statistically significant additional sales volume.
- Limiting discounts to 20% or below can maximize sales volume while protecting profitability.

K-Means Clustering Identified Five Customer Segments with Distinct Purchasing Behaviors



Customer Segment	Key Characteristics
High-Value Customers	Highest sales, profit, and purchase frequency
Loyal Customers	Frequent purchases with strong revenue contribution
Regular Customers	Moderate purchasing activity and low purchase frequency
Discount-Dependent Customers	Heavy discount usage with negative profitability
At-Risk Customers	Long inactivity and declining purchase activity

Heavy Discounting Increased Revenue, but Failed to Generate Positive Profit



Consumer customers contributed over half of sales, yet all customer types recorded negative profit, highlighting the limited effectiveness of excessive discount strategies.

EXCECUTIVE SUMMARY

Area	Key Findings
Business Performance	Profit exhibited recurring seasonal declines despite steady sales growth.
Product Performance	Technology drove profitability, whereas Tables remained the largest source of losses.
Profitability Driver	Deep discounts (>20%) significantly reduced profit without increasing sales volume.
Market Performance	APAC was the strongest market overall, although July performance weakened due to declining demand in key sub-categories.
Customer Intelligence	Five customer segments were identified, enabling targeted retention and profitability strategies.

STRATEGIC BUSINESS RECOMMENDATIONS

Four strategic initiatives to improve profitability and drive sustainable business growth

PRICING OPTIMIZATION

- Limit discount to < 20% to protect profitability
- Reduce discounts on Tables and other low margin products

PRODUCT OPTIMIZATION

- Prioritize high-margin products such as Phones and Copiers
- Review pricing and assortment of underperforming sub-categories (e.g., Tables)

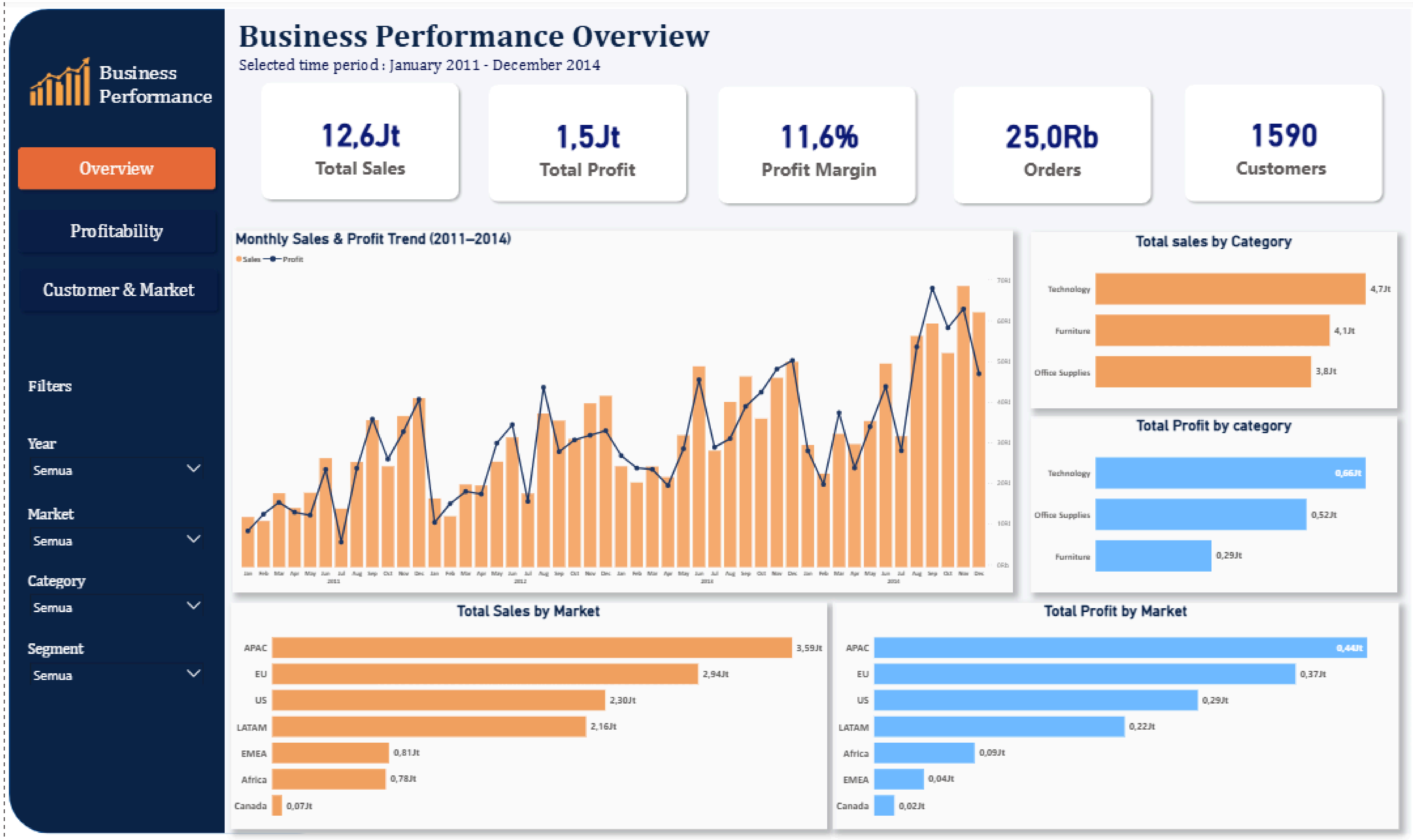
CUSTOMER STRATEGY

- Retain High Value Customers through loyality programs and personalized offers
- Reactive At Risk customers with targeted win back campaigns and incentives

MARKET STRATEGY

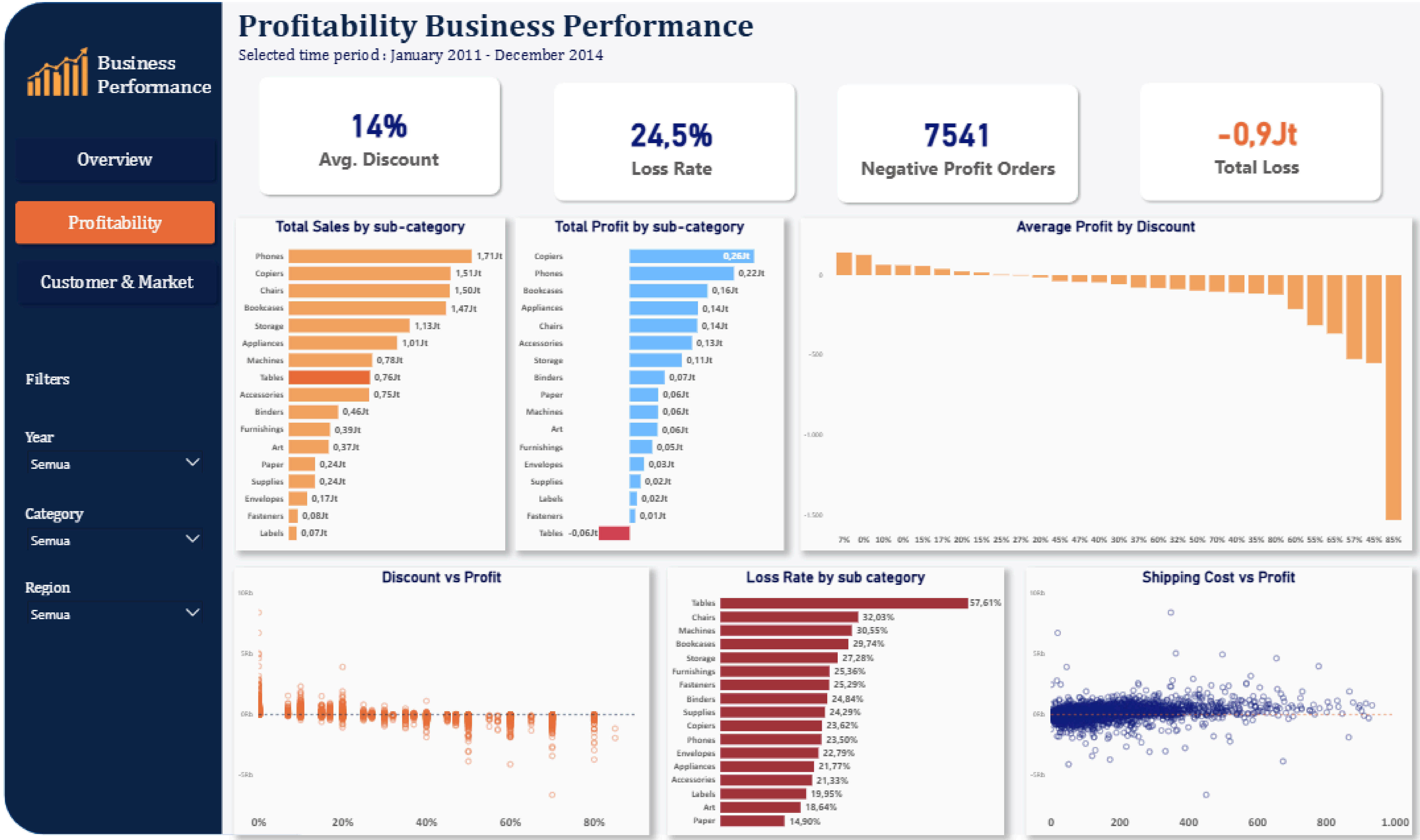
- Focus investment on APAC, the highest performing Market
- Replicate Canada’s low-discount, high-margin model where applicable
- Prepare targeted campaigns and inventory planning ahead of recurring July slowdown

INTERACTIVE DASHBOARD



Link Dashboard

INTERACTIVE DASHBOARD



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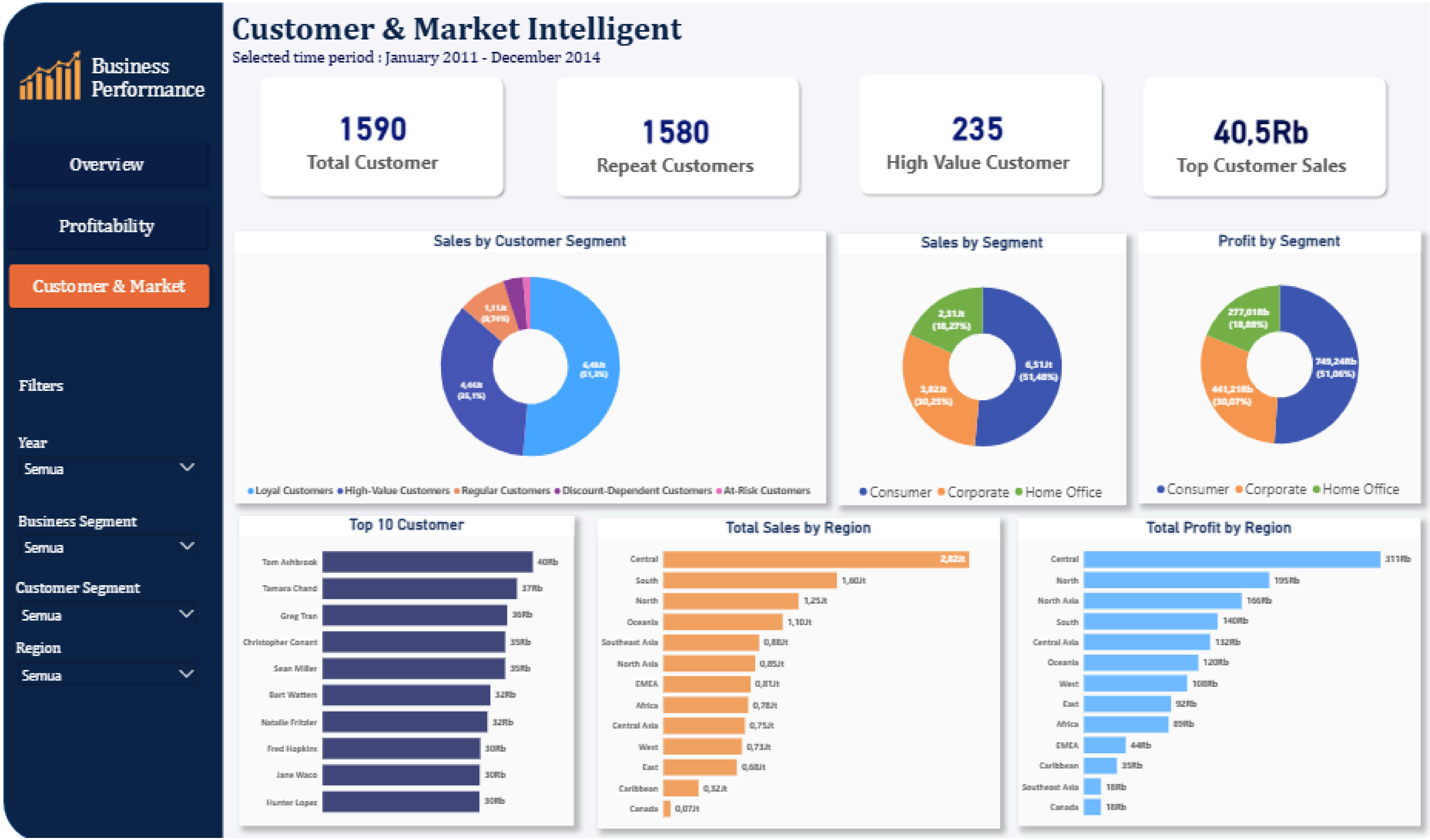

CONTACT ME

Get in touch

Call me
+62895351100201

Email
jessicaagnesiat@gmail.com

LinkedIn
jessicaagnesiat